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Do I Support That It's Good or Oppose That It's Bad? The Effect of Support-Oppose Framing on Attitude Sharing

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The rise of social media has led to unprecedented opportunities for individuals to share, or express, their attitudes on social and political issues. What factors affect whether individuals choose to share? This research identifies a novel determinant of attitude sharing—*support-oppose framing*, defined as whether individuals think of their own attitude in terms of what they support or what they oppose. Support-oppose framing is distinct from attitude valence, as the same attitude can be framed in terms of support (e.g., I support that this policy is bad) or opposition (e.g., I oppose that this policy is good). Seven experiments, two correlational studies, and one field study provide evidence for a support-oppose framing effect, whereby individuals are more likely to share attitudes framed in terms of positions they support rather than positions they oppose. This effect occurs via two pathways. In the first, support-framed attitudes are viewed as more value expressive, which facilitates greater attitude sharing. In the second, support-framed attitudes are believed to promote more positive impressions, which also leads to greater sharing. This effect is attenuated when individuals' typical impression-management goals are relaxed.

Keywords: attitudes, attitude sharing, opinion expression, advocacy, framing effects

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The proliferation of social media has led to unprecedented opportunities for individuals to share, or express, their attitudes on social and political issues. According to recent estimates, people post approximately 500 million tweets (Twitter, 2020), 2 million posts on Reddit (Reddit, 2015), and 4 billion pieces of content on Facebook (Protalinski, 2011) every day. Indeed, some estimates suggest that 94% of social media users share content online, with 42% of them sharing daily (Herhold, 2019). Not surprisingly, in recent years social psychologists have become increasingly interested in understanding the factors that drive attitude sharing—that is, in identifying the variables that lead people to express their attitudes and opinions (e.g., Akhtar & Wheeler, 2016; Berger, 2014; Cheatham & Tormala, 2015, 2017; Krosnick et al., 1993; Morrison & Miller, 2008). Research to date has identified individual differences (e.g., Akhtar & Wheeler, 2016; Rios et al., 2012), audience or target factors (e.g., Bechler et al., 2020; Dubois et al., 2016), and attitudinal attributes (e.g., Cheatham & Tormala, 2015; 2017; Gal & Rucker, 2010; Krosnick et al., 1993; Visser et al., 2003) as critical contributors.

The current research investigates a novel determinant of people's likelihood of sharing their views. Specifically, we explore the role of *support-oppose framing*, defined as whether people think of their attitudes in terms of what they support or what they oppose. Consider an individual who is reflecting on their attitude toward guns. This individual could be favorably inclined toward guns, and could think of their attitude in terms of what they support (e.g., I support allowing guns) or what they oppose (e.g., I oppose banning guns). Another individual could be negatively inclined toward guns, and think of *their* attitude in terms of what they support (e.g., I support banning guns) or what they oppose (e.g., I oppose allowing guns). Support-oppose framing is conceptually distinct from the attitude (or attitude valence) itself.

We ask whether variation in support-oppose framing, controlling for one's underlying attitude, affects people's likelihood of sharing their attitudes. And if so, how? To address these questions, we conducted a field study, two correlational studies, and a series of experiments. The results converge to show that people are more likely to share their attitudes when those attitudes are framed in terms of positions they support rather than positions they oppose. This effect is driven by two interaction goals: value expression and impression management. In the *value-expression pathway*, individuals feel that sharing their support for ideas and positions is more expressive of their core values (e.g., more representative of who they are) than is sharing their opposition to ideas and

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positions. In the *impression-management pathway*, individuals believe that expressions of support lead others (e.g., message recipients) to form more positive impressions of them than do expressions of opposition. The current studies demonstrate this effect across a variety of stimuli including social issues, policies, and brands, and illuminate the critical mediating roles of value expression and impression management.

Support-Oppose Framing and Sharing

Past research suggests that a variety of factors can influence people's decisions to share their views with others. Most germane to the current concerns, research documents the impact of attitudinal attributes on sharing. For example, attitudes that are held with greater certainty, deemed more important, or considered more self-defining are more likely to be shared, expressed, or talked about than their low certainty, low importance, and nondefining counterparts (Akhtar & Wheeler, 2016; Cheatham & Tormala, 2015; 2017; Krosnick et al., 1993; Visser et al., 2003; Zunick et al., 2017). We postulate that regardless of the attitudes people hold, the framing of those attitudes also affects whether they choose to share them.

In particular, we explore the role of *support-oppose framing*, defined as whether people think of their attitudes in terms of what they support or what they oppose. A support-framed attitude reflects what an individual supports or believes; it represents a viewpoint that the individual agrees with or endorses. An oppose-framed attitude reflects what an individual opposes or does not believe; it represents a viewpoint that the individual disagrees with or rejects. We submit that even after controlling for people's attitudes—that is, whether their underlying position on a topic is positive or negative—the framing of those attitudes in support or oppose terms can shape sharing decisions. Our central hypothesis is that people are more likely to share their views when they think of those views in terms of what they support rather than what they oppose.

The idea that framing can affect attitude-relevant outcomes has a rich history (e.g., Cesario & Higgins, 2008; Kahneman & Tversky, 1979; Kühberger, 1995; Lee & Aaker, 2004; Mayer & Tormala, 2010; McElroy & Seta, 2003). Importantly, though, the vast majority of prior work on this topic explores the effects of issue or message framing on people's attitudes, judgments, and decisions (e.g., see Levin et al., 1998; see also Nelson et al., 1997). In work on *issue* framing, for instance, researchers have found that framing a politician's actions as saving 40% versus losing 60% of a city's jobs can affect attitudes toward that politician (Boydstun et al., 2019), and that framing protests as a matter of free speech versus public safety can affect attitudes toward those protests (e.g., Chong & Druckman, 2007). Relatedly, research has revealed that the framing of survey questions (e.g., asking whether something should be forbidden or not allowed) can affect people's attitudes toward the queried issues (e.g., Holleman, 1999; Rugg, 1941). In addition to issue framing, extensive research has been conducted on *message* framing. The dominant finding in this literature is that matching message frames to recipients' psychological states or orientations can influence the effectiveness of persuasive messages. For example, matching message frames to a person's self-schema (Wheeler et al., 2005), motivational orientation (e.g., approach vs. avoidance; Updegraff et al., 2007), regulatory focus (e.g., promotion vs. prevention; Cesario & Higgins, 2008; Lee & Aaker, 2004), cognitive versus affective orientation (Mayer & Tormala, 2010), social identity (Mackie et al., 1990), or self-guide orientation (e.g., ideal vs. ought self; Evans & Petty, 2003) can affect persuasion by increasing message processing or eliciting feelings of fit or fluency.

Despite extensive work on issue and message framing, scant research has explored the effect of *attitude* framing on attitude-relevant outcomes. That is, very little attention has been paid to the notion that people can frame their attitudes in various ways, and that these frames are consequential. In one notable exception, Bizer et al. (2011; see also Bizer & Petty, 2005) uncovered what they called a "valence framing" effect, whereby framing a political attitude as being in favor of a liked candidate reduced resistance to persuasion relative to framing the attitude as being against their opponent (i.e., a disliked candidate). This effect was mediated by attitude certainty: Opposing a disliked candidate boosted certainty relative to supporting a liked candidate, which increased resistance to subsequent persuasion. However, there are important caveats to these findings. First, Bizer et al. (2011) did not examine sharing decisions, which are the focus of the current research. Second, if anything, the Bizer et al. (2011) findings would point to the opposite prediction to the one advanced here. Oppose framing boosted attitude certainty in the Bizer et al. (2011) work, and certainty generally promotes attitude sharing (e.g., Cheatham & Tormala, 2015; 2017). Thus, the Bizer et al. (2011) findings imply that oppose framing should trigger more sharing than support framing. We predict the opposite.

As a note, although the research differs in multiple ways, the general construct of support-oppose framing closely parallels what Bizer et al. (2011) referred to as "valence framing." In each case, the focus is on how people with the same underlying attitude toward an issue or entity can frame that attitude in terms of the position, or side, they support or oppose. In the current research we adopt the term "support-oppose framing" for two reasons. First, it speaks directly to the focal dichotomy between support- and oppose-framed attitudes. Second, we distinguish support-oppose framing from attitude valence, and view the term "valence framing" as potentially confusing or ambiguous regarding the relationship between an attitude's valence and frame. Indeed, others have argued that "valence framing" is a multifaceted construct that has been used to refer to different phenomena in past research (Levin et al., 1998; see also Sparks, 2020). Thus, although support-oppose framing is essentially equivalent to Bizer et al.'s (2011) use of "valence framing," for clarity we adopt the term "support-oppose framing" in this research.

Motives for Sharing

What motivates people to share their attitudes, and how are these motives shaped by support-oppose framing? To answer these questions, we draw on the attitude function literature. Functional theories of attitudes generally suggest that attitudes serve five primary functions: a knowledge function, a utilitarian (or object-appraisal) function, an ego-defense function, a value-expression function, and a social-adjustment function (e.g., Katz, 1960; Shavitt, 1989; Smith et al., 1956). The first three of these functions would seem to explain why people *hold* attitudes, but not necessarily why they choose to share their attitudes with others. For

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example, the knowledge function suggests that attitudes help individuals understand, organize, and predict the world around them. Attitudes can provide this understanding and give the world a sense of predictability even if they are not expressed to others. The utilitarian (or object-appraisal) function suggests that attitudes allow individuals to store positive and negative associations—for instance, information about reward and punishments—that guide approach and avoidance behaviors and interactions with attitude objects. For example, an individual's attitude toward pizza is tied in part to the reward and punishments associated with consuming pizza, which guides people to approach or avoid pizza. Like the knowledge function, the utilitarian function is not directly tied to attitude sharing; positive (negative) attitudes should direct approach (avoidance) behavior even if they have not been expressed to others. Finally, the ego-defense function suggests that attitudes can be used to protect the self-concept—for example, by derogating outgroups in response to threat (e.g., Crocker et al., 1987). Because protecting the self-concept can be a private or internal process, the ego-defense function might also be less important as a determinant of attitude sharing.

The remaining attitude functions—value expression and social adjustment—have often been discussed together as a social identity function, and they are thought to guide the impact of people's attitudes on their interactions and social relationships. For example, the value-expression function suggests that attitudes allow people to express their core values and self-concepts. People view some attitudes as symbolic of the self (Shavitt, 1989), expressing these attitudes allows people to declare who they are and what they stand for. The social-adjustment function suggests that attitudes can be leveraged to facilitate social relations, gain approval from others, and aid in impression management. The logic is that expressing attitudes can create social bonds or positive impressions when doing so facilitates social approval (Smith et al., 1956) or fosters a sense of common ground (Byrne, 1969). Given that the value-expression and social-adjustment functions are inherently social, they might play critical roles in attitude sharing.

We leverage prior theories on attitude function and predict that when people consider sharing their attitudes with others, they are driven by two primary goals: the goal to express their values and the goal to promote positive impressions. In other words, we submit that two main interaction goals—value expression and impression management—drive attitude sharing. Indeed, it is well-established that interaction goals play a fundamental role in guiding people's interactions with others (Rusbult & Van Lange, 2003), and value expression (desiring to present and view oneself as congruent with one's values and ideal self-image) and impression management (desiring for others to have a favorable impression of oneself) have emerged as two central goals that drive a wide range of social interactions and behaviors (see Baumeister, 1982). We unpack each goal below and outline its hypothesized role in sharing.

Value-Expression Goal

One major interaction goal is value expression. People are motivated to act and portray themselves as aligned with their ideal selves and values to affirm that they are who they want to be (Baumeister, 1982; Higgins, 1987). Given that values are a fundamental aspect of the self-concept (Hitlin, 2003), it follows that individuals should be motivated to express their core values when interacting with others. As noted, this proposition aligns with the attitude function literature, which suggests that individuals derive satisfaction from holding and expressing attitudes that reflect their personal values and self-concepts (e.g., Katz, 1960; Shavitt, 1989). Work in consumer psychology also suggests that a key motive for sharing information (e.g., about a recent purchase) is to express who one is (Berger, 2014; Wojnicki & Godes, 2017). Thus, when an attitude is viewed as more value expressive, people should be more likely to share it.

We hypothesize that support-framed attitudes are viewed as more value expressive than oppose-framed attitudes. We base this hypothesis on prior work on self-definition, which suggests that people generally define themselves more by who they *are* (e.g., liberal) than by who they are not (e.g., not conservative; Zhong et al., 2008). The preference for identities framed in terms of who one is, rather than is not, is also suggested by optimal distinctiveness theory (Brewer, 1991). According to this theory, social identity is driven by two fundamental and competing human needs: the need to be unique or distinct from others and the need to belong or feel included. Representing the self in terms of who one *is* meets both of these needs (belonging to an ingroup and being distinct from an outgroup), whereas representing the self in terms of who one *is not* meets only one of these needs (being distinct from an outgroup). We propose that a similar process occurs when people reflect on and express their attitudes. Reflecting on support frames an attitude in terms of who one is or what one believes; reflecting on opposition frames the attitude in terms of who one is not or what one does not believe. We posit that the former is considered more value expressive than the latter.

Impression-Management Goal

A second major interaction goal involves impression management. An extensive literature suggests that the desire to be liked is a fundamental human motivation (e.g., Ames et al., 2011; Baumeister, 1982; Jones & Pittman, 1982). People engage in a wide range of behaviors (e.g., altruism, conformity, and ingratiation) to give others more favorable impressions of them (Baumeister, 1982; Jones & Pittman, 1982). Thus, we argue that people should be more likely to share when they believe that doing so will lead others (i.e., those with whom they are interacting) to form more favorable impressions of them. As noted, this reasoning is also consistent with the attitude-function literature, which suggests that people use attitude expression to project social identities that will elicit approval from others (Chung & Darke, 2006; Hennig-Thurau et al., 2004; Hughes, 2005; Shavitt, 1989; Sundaram et al., 1998; Wojnicki & Godes, 2017).

We hypothesize that people believe that support-framed attitudes will better serve this impression-management goal than oppose-framed attitudes. One reason is that support-framed positions essentially represent agreement (we support what we agree with), whereas oppose-framed positions constitute disagreement (we oppose what we disagree with). Past research on interpersonal liking reveals that individuals like others who agree with them more than others who do not (Berscheid & Hatfield, 1969; Byrne, 1969). Furthermore, individuals seem to capitalize on this tendency and voice agreement rather than disagreement (Berscheid & Hatfield, 1969; Newcomb, 1953), particularly when they have a

strong desire to be liked (Jellison & Gentry, 1978; Schlenker, 1980). If people generalize this tendency and extend their intuition about voicing agreement with others to voicing agreement of any kind—regardless of others' views—this lay belief could manifest as generally increased expression of support-framed (agreement-focused) attitudes relative to oppose-framed (disagreement-focused) attitudes, even when it does not offer an effective impression-management tactic. To assess this lay belief, we asked 200 participants on Amazon's Mechanical Turk which would make others view them more positively: to advocate against something, to advocate in support of something, or neither. Fifty-four percent reported that advocating in favor would be a better strategy and only 10% indicated that advocating against would be better. Thus, it appears that most people link expressions of support with likeability. We examine the implications of this intuition for attitude sharing.

Attitude Valence

Although we could identify no prior studies investigating the effect of support-oppose framing on sharing decisions, some research suggests that attitude valence—that is, whether the underlying attitude itself is positive or negative—predicts sharing-related outcomes. In general, positivity appears to be associated with greater sharing than negativity. For example, consumer research suggests that people write more positive than negative reviews (Chevalier & Mayzlin, 2006; East et al., 2007) and share more positive than negative news (Berger & Milkman, 2012; Tesser & Rosen, 1975). This tendency is thought to stem at least in part from the perception that sharing positive content increases one's likeability (Berger, 2014). There are exceptions to this finding, though, and some have argued that people share positive or negative content depending on which they think will make them look better in the moment (De Angelis et al., 2012). Further evidence comes from research on self-defining attitudes. Zunick et al. (2017) found that people are more likely to advocate on behalf of their own views (e.g., argue to persuade others to one's side) when those views feel self-defining, and in general attitudes feel more self-defining as they become more favorable. For example, people with positive rather than negative attitudes toward environmentalism feel more defined by their views, and are more likely to try to persuade others.

Attitude valence and support-oppose framing are related constructs and certainly can move together. For example, in the case of an election with two candidates (see Bizer et al., 2011), supporting one candidate and having a more positive attitude toward that candidate necessarily co-occur. Thus, research pointing to a positive correlation between attitude valence and sharing-related outcomes hints at the tenability of our hypothesis regarding support-oppose framing and sharing. Importantly, though, valence and framing are conceptually distinct and empirically separable. In the gun example, a person with a positive attitude toward guns could frame their attitude in terms of support (e.g., "I support allowing guns") or opposition (e.g., "I oppose banning guns"). Likewise, a person with a negative attitude could frame their attitude in terms of support (e.g., "I support banning guns") or opposition (e.g., "I oppose allowing guns"). In neither case does the frame shift the individual's core attitude toward guns (see Table 1 for additional examples). Thus, past research on valence and sharing is suggestive but inconclusive with respect to the current research question. If support framing does promote sharing, this finding would highlight a novel trigger of attitude expression, deepen our understanding of attitude framing effects, and offer practical insight to influencers seeking to stimulate advocacy or motivate others to express their views.

Overview

In sum, we hypothesize two interaction goals—value expression and impression management—that mediate the effect of support-oppose framing on attitude sharing. In the first pathway, we contend that people are more likely to share support- rather than oppose-framed attitudes because they believe that the positions they support are more representative and reflective of their values than the positions they oppose. In the second pathway, we posit that people are more likely to share support- rather than oppose-framed attitudes because they believe doing so will cause others to form more favorable impressions of them. If obtained, these results would make significant contributions to our understanding of attitudes. For example, they would deepen our insight into the factors that lead people to express their views, contributing to a rapidly expanding literature on opinion expression and attitudinal advocacy (Tormala & Rucker, in press). In addition, they would extend theorizing on attitude function and interaction goals, expanding our understanding of the motives that drive attitude expression. Finally, they would underscore support-oppose framing as an important psychological construct, thus broadening the scope of attitudes research beyond valence and sounding a call for research on the factors that prompt people to think of their attitudes in terms of support or opposition.

Table 1
Examples of Positive and Negative Attitudes Expressed Using Support and Oppose Framing

Attitude object	Positive attitude	Negative attitude
	Support frame	
guns	I support allowing guns.	I support banning guns.
same-sex marriage	I'm in favor of making same-sex marriage legal.	I'm in favor of making same-sex marriage illegal.
Barack Obama	I agree that Obama had a positive effect on the U.S.	I agree that Obama had a negative effect on the U.S.
	Oppose frame	
guns	I oppose banning guns.	I oppose allowing guns.
same-sex marriage	I'm against making same-sex marriage illegal.	I'm against making same-sex marriage legal.
Barack Obama	I disagree that Obama had a negative effect on the U.S.	I disagree that Obama had a positive effect on the U.S.

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Ten studies test our predictions. Study 1 provides initial field evidence of the relationship between support-oppose framing and sharing using Twitter data. Studies 2a and 2b explore the relationship across ten different issues—presented in terms of attitude or position statements—and disentangles support-oppose framing and valence by varying statement valence. Study 3 offers causal evidence for the support-oppose framing effect in a laboratory experiment. To elucidate the mechanisms underlying this effect, Study 4 measures how value expression and perceived impressions differ across support- and oppose-framed attitudes. Study 5 triangulates in on psychological process and moderates the effect by manipulating desired impression. Supplemental Studies 1–4 provide additional evidence for the proposed mechanism and explore alternative explanations for our effect. Finally, we present a meta-analysis assessing the relative effects of support-oppose framing and valence across all studies.

Before presenting our studies, it is worth noting a few methodological points. First, across studies, we investigated support-oppose framing in the context of bipolar attitudes and potentially polarizing issues—that is, in contexts in which supporting an issue or idea would be the converse of opposing it, and in which people can be quite divided. This focus was based on two factors: (a) These contexts are increasingly prevalent today. We live in a world of ever-growing division and political polarization (Pew Research Center, 2017), and it is more important than ever to understand the factors that drive attitude sharing in divisive settings involving contentious issues. (b) Focusing on bipolar contexts permitted tighter study designs, in which supporting one position could be logically equivalent to opposing another. Second, across studies, our interest was in participants' sharing decisions—that is, their desire and decision to express their views—rather than in the *extent* of their sharing (e.g., the amount they would say or write when given an opportunity to do so). Thus, our primary measures and analyses involved sharing intentions. For exploratory purposes, however, some studies did invite participants to actually express their views (i.e., write about them), and we analyzed the extent of sharing (i.e., how much participants wrote) as an exploratory analysis in those studies. For these analyses, we refer readers to the online supplemental materials. Finally, across studies, we made several efforts to maximize power, using power calculations, large samples, and repeated measures where appropriate for a total of over 70,000 observations. We also employed attention and consistency checks to reduce noise in the data. All methods and hypotheses were preregistered and all materials and data are available on OSF (https://osf.io/p3q9f).

Study 1: Twitter

To offer an initial assessment of the relationship between support-oppose framing and sharing, we conducted a field study using the social media platform, Twitter. In particular, we examined Twitter users' posts to investigate whether individuals expressed more support-framed attitudes than oppose-framed attitudes in this natural environment.

Method

We collected data through the Twitter Application Programing Interface (API) using the rtweet package for R (Kearney, 2016). Tweets were selected based on two criteria: topic and support/oppose framing. First, we limited tweets to those including the words "policy," "law," or "legislation." Using these words allowed us to focus on topics of interest and avoid instances of "support" that are not germane to the present research (e.g., requests for support, such as "please support my GoFundMe"). In addition, using these general words (rather than more specific keywords, such as "gun" or "immigration") allowed us to make inferences about support and opposition broadly, rather than limiting our conclusion to a particular issue or group of issues.

The second criterion was the presence of words reflecting support or oppose framing. Tweets containing the phrases "I support," "I am in favor of," or "I agree" were categorized as support tweets; tweets containing the phrases "I oppose," "I am against," or "I disagree" were categorized as oppose tweets. Full phrases were used to minimize the possibility of including tweets that were ambiguous in their support versus opposition. For example, a tweet urging others, "Do not support this policy," is ambiguous in whether it should be labeled as support framing (based on the use of the word "support") or oppose framing (based on the position of the speaker). By including only tweets in which people discussed their personal views in terms of support or opposition, we were able to conduct a focused test of our hypothesis.

The final sample included 63,069 tweets from a total of 44,085 users and included all relevant tweets available via the Twitter API for the month in which we collected data. One month was used as the time span (rather than a shorter time period) to ensure a dataset that would be minimally biased by a single event or tweet. The data collection strategy (including data collection window) and subsequent analyses were preregistered on AsPredicted.[1]

Results

Tweets were coded based on whether they used support language, oppose language, or both. Results indicated that Twitter users were significantly more likely to tweet using support rather than oppose language, $N_{\text{support}} = 50{,}037$, $N_{\text{oppose}} = 1{,}087$; $\chi^2(1) = 46{,}868.45$, $p < .001$. Including tweets in which individuals used support, oppose, or both support and oppose language ($N_{\text{support}} = 50{,}037$, $N_{\text{oppose}} = 1{,}087$; $N_{\text{both}} = 11{,}945$), this significant difference persisted, $\chi^2(2) = 62{,}867.64$, $p < .001$.[2] A Bonferroni-corrected posthoc test revealed a significant difference for every pairwise comparison, $p < .001$. In addition to the number of tweets, we also analyzed the prevalence of support ("support," "agree," and "favor") versus oppose ("oppose," "disagree," and "against") words on Twitter. We found a consistent and significant effect, such that more support words were used than oppose words, $t(63{,}068) = 200.52$, $p < .001$. Full details for this latter analysis are included in the online supplemental materials.

If people are more likely to share their attitudes when those attitudes are framed in terms of support rather than opposition, then it follows that support-framed tweets might generally receive more

[1] https://aspredicted.org/blind.php?x=hp8uh6

[2] Because most users only had a single tweet in the dataset, mixed models with a random effect for participant to account for repeated measures at the participant level are overspecified and fail to converge. See online supplemental materials for an alternative analysis using one tweet per user (i.e., no repeated measures).

retweets than oppose-framed tweets. Our findings were consistent with this proposition: Tweets that used support language were retweeted an average of 624 times each, tweets that used oppose language were retweeted an average of 28 times each, and tweets containing both support and oppose language were retweeted an average of 542 times each, $F(2, 63{,}066) = 162.29$, $p < .001$. The key difference between support and opposition retweeting persisted ($M_{\text{both}} = 3.61$, $M_{\text{oppose}} = 1.54$; $M_{\text{support}} = 3.25$, $F(2, 63{,}066) = 232.86$, $p < .001$) after log transforming the data to normalize and account for outliers (Gelman & Hill, 2007).

Is it possible that the prevalence of support-framed tweets was driven by valence (i.e., the desire to post positive tweets) rather than framing per se? To answer this question and determine whether valence and framing were orthogonal in the data, we used the Evaluative Lexicon 2.0 (Rocklage et al., 2018) in an exploratory analysis to classify tweets based on their use of positive language, negative language, or both. The Evaluative Lexicon uses a dictionary of over 1,500 extensively validated adjectives to analyze the content of written text for valence—searching, for instance, for words such as "amazing" and "terrible" to identify positive and negative evaluations, respectively. For our analysis, we included tweets if they were unique (i.e., not retweets) and contained both valenced language and support-oppose language (N = 27,803). If the prevalence of support-framed tweets is driven by valence, we would expect to see a greater preponderance of support tweets using positive language, and oppose tweets using negative language. Instead, we found that most tweets (55%) contained negative language, and the framing pattern held regardless of valence (see Table 2), underscoring the separability of these two constructs. Thus, the framing effect appeared to be independent of valence.

Discussion

In Study 1, we found that people were more likely to use support-related than oppose-related language in a real-world setting, by assessing actual behavior across thousands of individuals and tweets. Moreover, by leveraging text analysis we were able to separate support and oppose framing from valence. We found that support language did not necessarily imply positivity, nor did oppose language imply negativity. Indeed, among every category of tweets (including those containing support language), negativity was more prevalent than positivity. That people were more likely to use negative than positive language in their tweets is consistent with a potential negativity bias (Baumeister et al., 2001), which can be particularly prominent on social media (Pew Research Center, 2012, 2015). Most crucial to our concerns, within tweets containing positive language, negative language, or both (ambivalence), support framing was more prevalent than oppose framing.

Table 2
Absolute Number of Tweets as a Function of Valence and Support-Oppose Framing

	Language valence		
	Positive	Negative	Both (ambivalent)
Support framing	6,773	12,698	3,237
Oppose framing	124	335	70
Both	1,795	2,306	465

Although testing our hypothesis across many people in a natural environment (i.e., Twitter) has advantages, Study 1 is not without limitation. One limitation is that due to the scale of Study 1, we relied on text analysis to code support-oppose framing and valence. To code framing, we relied on a preselected set of support- and oppose-focused words and phrases (support = support, agree, in favor of; oppose = oppose, disagree, against), which might not have captured all the ways people express support and opposition. It is possible that a wider range of words and phrases would produce a different pattern of results. Another limitation is that we used a linguistic heuristic (tweets containing the words "policy," "law," or "legislation") to ensure that support- and oppose-related words were being used with the intended meaning. Without hand-coding the tweets, it is impossible to be sure that every tweet did express support or opposition—that is, share support- and oppose-framed attitudes—as intended. Finally, because we also used text analysis to categorize valence, we cannot be certain that our valence categories mapped perfectly onto positive and negative attitudes. To address these concerns, subsequent studies were conducted in more controlled settings.

Studies 2a and 2b: Multi-Issue Correlations

Study 2 had several objectives. First, we sought to more directly assess attitude sharing across a variety of topics. To this end, we measured participants' degree of support or opposition and corresponding sharing intentions across social issues (Study 2a), political figures (Study 2a), brands (Studies 2a and 2b), and company policies (Study 2b). Second, we sought to explore the shape of the relationship between degree of support versus opposition and sharing intentions. We expected sharing intentions to rise as people's degree of support or opposition grew more extreme (consistent with work suggesting that people are more likely to talk about more extreme attitudes; Krosnick et al., 1993), but also expected support to be associated with greater sharing intentions than opposition.

Finally, in Study 2 we aimed to more directly disentangle support-oppose framing from attitude valence. To do so, we asked participants to indicate their degree of support or opposition to a series of statements that varied in valence. For example, one statement read, "Uber has had a positive effect on America." Participants who supported this statement would have a positive attitude toward Uber and participants who opposed this statement would have a negative attitude toward Uber. Another statement read, "Facebook has had a negative effect on America." Here, supporting the statement implied a negative attitude, whereas opposing it implied a positive attitude. By varying statement valence across topics, we were able to explore the independent roles of support-oppose framing and attitude valence in predicting sharing intentions.

Method

Based on pilot testing, we set a target sample size of 400 participants for both Study 2a and Study 2b. In Study 2a, 401 participants completed our survey on Amazon Mechanical Turk. A total of 354 (43.79% female, $M_{\text{age}} = 34.89$) were included in the final analysis, after exclusions as specified in our

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preregistration.[3] In Study 2b, 400 participants completed our survey on Amazon Mechanical Turk. A total of 355 (45.35% female, M_{age} = 36.33) were included in the final analysis, after exclusions as specified in our preregistration.[4] In both studies, participants were asked about their attitudes toward a series of randomly ordered statements regarding social and political issues and policies (see Table 3). For statements about specific policies, participants were also given a description of the policy in question.

We varied the statements on multiple dimensions (e.g., positive vs. negative, social/political issue vs. company policy, supported by liberals vs. conservatives vs. apolitical) to ensure that "support" was not consistently associated with positive underlying attitudes (nor "oppose" consistently associated with negative underlying attitudes) or consistently chosen by a single ideological group. After each statement, participants were asked to report their support or opposition to the statement on a scale ranging from 0 (*strongly oppose*) to 100 (*strongly support*). This rating was used to naturally activate a support or opposition focus for each participant (i.e., to induce participants to think of their attitudes in support or oppose terms), with lower scores indicating oppose framing and higher scores indicating support framing. Because the statements varied in valence and position, higher (lower) scores indicated more support (opposition), but not necessarily more positive (negative) underlying attitudes.

After indicating support or opposition for each issue, participants completed a measure of sharing intentions adopted from Cheatham and Tormala (2015). Specifically, participants answered three questions assessing their likelihood of sharing their views with a friend or family member who disagreed with them, an acquaintance who disagreed with them, or a stranger who disagreed with them (e.g., "How likely would you be to share your views on this topic with your friends or family who disagree with you on this issue?" 1 = *not at all likely;* 7 = *extremely likely*). Responses were consistent (Study 2a α = .88, Study 2b α =. 90), and were averaged to form a single sharing index.[5] Note that in Studies 2a and 2b (see also Study 3), we focused the sharing items on disagreeing others. We focused on disagreeing others to shed light on the factors that guide attitude sharing in divisive contexts, which increasingly characterize our social and political world (Pew Research Center, 2017) and in which sharing might be more limited.

Table 3
List of Topics Used in Studies 2a and 2b

Study 2a
Barack Obama had a positive effect on America.
Walmart has had a negative effect on America.
Abortions should be illegal in America.
Same-sex marriage should be legal in America.
Study 2b
Uber has had a positive effect on America.
Facebook has had a negative effect on America.
Walmart has had a negative effect on America.
Bank of America's new policy requiring a minimum balance to avoid fees is bad for its customers.
Starbucks' new policy removing plastic drinking straws from their stores is good for America.
Google's policy allowing employees to listen to recordings to improve service is bad for America.
Coca-Cola's new policy paying for employees' gender assignment surgeries is bad for America.

Results

For our main analysis, we used mixed-models treating the linear and quadratic terms for participants' opposition or support as fixed-effects and participant and topic as random-effects.[6] We calculated *t*-values and *p*-values using lme4 and lmerTest, respectively (Bates et al., 2015; Kuznetsova et al., 2015). Degrees of freedom for *t*-statistics are not reported due to uncertainty surrounding their calculation in mixed models (e.g., Baayen et al., 2008). Degrees of freedom for the reported *F*-statistics exceed the total *N* in these analyses because using multiple topics produced multiple observations per participant.

In both Study 2a and Study 2b, participants expressed greater sharing intentions for statements they supported than for statements they opposed. As illustrated in Figure 1 (top panels), participants reported greater intentions to share their views as those views became increasingly reflective of support rather than opposition, Study 2a: ß = 8.06, t = 11.30, $p < .001$; Study 2b: ß = 12.07, t = 17.00, $p < .001$. In addition, we found the predicted quadratic relationship such that participants with more extreme support or opposition were more likely to share their views than were participants with more moderate support or opposition, Study 2a: ß = 12.38, t = 15.83, $p < .001$; Study 2b: ß = 16.06, t = 21.50, $p < .001$. For an exploration of the framing effect by topic, we refer readers to the online supplemental materials.

Given our interest in comparing sharing intentions across issues or positions that participants support versus oppose, as a supplemental analysis we divided observations into three groups based on the position taken: opposed to the statement (opposition-support scale response < 50; Study 2a N = 530; Study 2b N = 770), neutral to the statement (opposition-support scale response = 50; Study 2a N = 61; Study 2b N = 131), and supportive of the statement (opposition-support scale response > 50; Study 2a N = 825; Study 2b N = 1584). Using mixed-models as above, we found a significant difference in sharing intentions across the three groups, Study 2a: $F(2, 1144) = 45.5$, $p < .001$; Study 2b: $F(2, 2202) = 90.80$, $p < .001$. As shown in Figure 1 (bottom panels), participants who supported the statements reported greater sharing intentions (Study 2a: M = 4.66, 95% CI [4.55, 4.78]; Study 2b: M = 4.42; 95% CI [4.34, 4.51]) than participants who opposed them (Study 2a: M = 4.00, 95% CI [3.85, 4.16], t = 6.67, $p < .001$, d = .37; Study 2b: M = 3.66; 95% CI [3.53, 3.79]), t = 10.81, $p <$

[3] Study 2a: https://aspredicted.org/blind.php?x=rc68ni

[4] Study 2b: http://aspredicted.org/blind.php?x=w59du9

[5] As noted, our primary interest in this research was in capturing the role of support-oppose framing in shaping people's sharing decisions. In some studies (but not all), we also preregistered and measured persuasion intentions. In these studies, sharing and persuasion intentions were significantly correlated and generally displayed similar results. For interested readers, the measures and data for persuasion intentions are available on OSF.

[6] There is some debate about whether mixed models should use the maximal random effects structure (Barr et al., 2013) or focus on using the random effects structure best supported by the data, to balance maximizing power and minimizing type I error (Matuschek et al., 2017). We included random intercepts for participant and topic, consistent with the approach recommended by Matuschek et al. (2017). Using random slopes instead of random intercepts (as recommended by Barr et al., 2013) yields equivalent results.

.001, $d = .44$. In addition, participants who opposed the statements reported greater sharing intentions (Study 2a: $M = 4.00$, 95% CI [3.85, 4.16]; Study 2b: $M = 3.66$; 95% CI [3.53, 3.79]) than participants who were neutral (Study 2a: $M = 2.80$, 95% CI [2.42, 3.17], $t = -4.93$, $p < .001$, $d = .67$; Study 2b: $M = 3.41$; 95% CI [3.10, 3.72]), $t = -4.08$, $p < .001$, $d = .14$.

Finally, we also explored attitude valence as a predictor of sharing using a mixed-model, with a random effect for participant and topic, as well as terms for support-oppose framing (support/oppose on the 100-point scale) and the interaction between attitude valence and support-oppose framing. Participants who supported positive statements (i.e., statements with "good," "positive," or "legal") or opposed negative statements (i.e., statements with "bad," "negative," or "illegal") were coded as having positive attitude valence. Participants who opposed positive statements or supported negative statements were coded as having negative attitude valence. Those who indicated neutral attitudes were excluded from this analysis. We found no interaction between attitude valence and support-oppose framing on sharing intentions (Study 2a: linear: ß = 1.53, $t = .84$, $p = .44$; quadratic: ß = 2.53, $t = 1.52$, $p = .13$; Study 2b: linear: ß = .15, $t = .05$, $p = .96$; quadratic: ß = .56, $t = .32$, $p = .75$). Because the interaction was not significant, we then explored a mixed model predicting sharing intentions, with fixed effects for the linear and quadratic terms for support-oppose framing and a linear term for attitude valence, as well as a random effect for participant (topic was omitted due to insufficient data; Brauer & Curtin, 2018). We found no effect of attitude valence on sharing intentions, Study 2a: ß = −.04, $t = -.95$, $p = .34$; Study 2b: ß = −.02, $t = -.40$, $p = .69$. Controlling for valence, both the linear, Study 2a: ß = 7.97, $t = 11.64$, $p < .001$; Study 2b: ß = 11.54, $t = 13.00$, $p < .001$, and quadratic, Study 2a: ß = 11.25, $t = 14.15$, $p < .001$; Study 2b: ß = 15.39, $t = 19.73$, $p < .001$, relationships between support-oppose framing and sharing intentions persisted.

Discussion

In Studies 2a and 2b, we found that people reported greater intentions to share their views after reporting support rather than opposition across a variety of topics spanning social and political issues, brands, and company policies. By varying the presentation of these topics as statements reflecting positive or negative attitudes, we were able to distinguish support-oppose framing from attitude valence and explore the independent contributions of each. We found that the relationship between support-oppose framing and sharing held even after controlling for the underlying valence of people's attitudes. Moreover, we found no main effect of attitude valence in this study, nor an interaction between valence and framing. This finding, paired with the potential negativity bias in sharing decisions in Study 1, raises the possibility that prior research suggesting a positive relationship between valence and sharing-type outcomes might be overstated. In general, inconsistency in the valence effect suggests that the role of valence in attitude sharing might be unreliable. There could be undiscovered moderators of this effect, or it could be that valence is not a primary driver. Perhaps prior work pointing to a positive relationship confounded positive valence with support framing, and when these are unconfounded the valence effect attenuates. We return to this issue in the General Discussion section.

Study 3: In-Lab Experiment

The primary aim of Study 3 was to provide causal evidence for the support-oppose framing effect. To this end, we recruited university students and staff to the laboratory for a study about online interactions, and randomly assigned them to a support- or oppose-framing condition. Support- and oppose-framing were manipulated by presenting participants with an issue statement that they supported (or agreed with) versus opposed (or disagreed with). To ensure that participants would find the topic personally important, we selected gun control as the attitude issue and assessed issue importance at the end of the survey.

Method

On the basis of a pilot test indicating a Cohen's d of .30, we set a target sample size of 250 participants, yielding 85% power to detect a medium effect size after allowing for 50 exclusions. In total, 240 university students and staff participated in a lab Study for monetary compensation. After exclusions specified in our preregistration,[7] 206 observations were used in our final analysis (64.08% female, $M_{age} = 24.83$). At the outset, participants were told that they would be interacting in an online chatroom to discuss a social issue with another participant. To ensure that participants would see others taking the survey at the same time, thereby bolstering realism, participants completed the survey in groups. For any sessions in which only a single participant was present, a confederate acted as another participant.

After providing a username to identify themselves, participants were asked to indicate their attitude on three filler topics (same-sex marriage, abortion, and immigration), as well as the target topic (gun control). For the target question, participants were asked, "Which of the following statements best characterizes your attitude toward gun control?" and given three options: "More gun control (i.e., stricter gun laws) would be beneficial to the U.S.," "More gun control (i.e., stricter gun laws) would be harmful to the U.S.," or "no opinion." If participants indicated no opinion, they were asked to choose which of the two statements was closer to their beliefs. This allowed us to retain more participants and maximize sample size and power.

Participants were then randomly assigned to one of two conditions. In the support-framing condition, participants were told that they would be asked to answer questions about the statement they supported—that is, the one they selected. In the oppose-framing condition, participants were told that they would be asked to answer questions about the statement they opposed—that is, the one they did not select. In both conditions, participants were asked about an attitude position that was consistent with their own view. In other words, a participant who indicated that more gun control would be beneficial was randomly assigned to answer questions about their support for the beneficial statement or their opposition to the harmful statement. A participant who indicated that more gun control would be harmful was randomly assigned to answer questions about their support for the harmful statement or their opposition to the beneficial statement. Thus, all participants answered questions about their own stance on gun control, but it

[7] http://aspredicted.org/blind.php?x=u8v4u3

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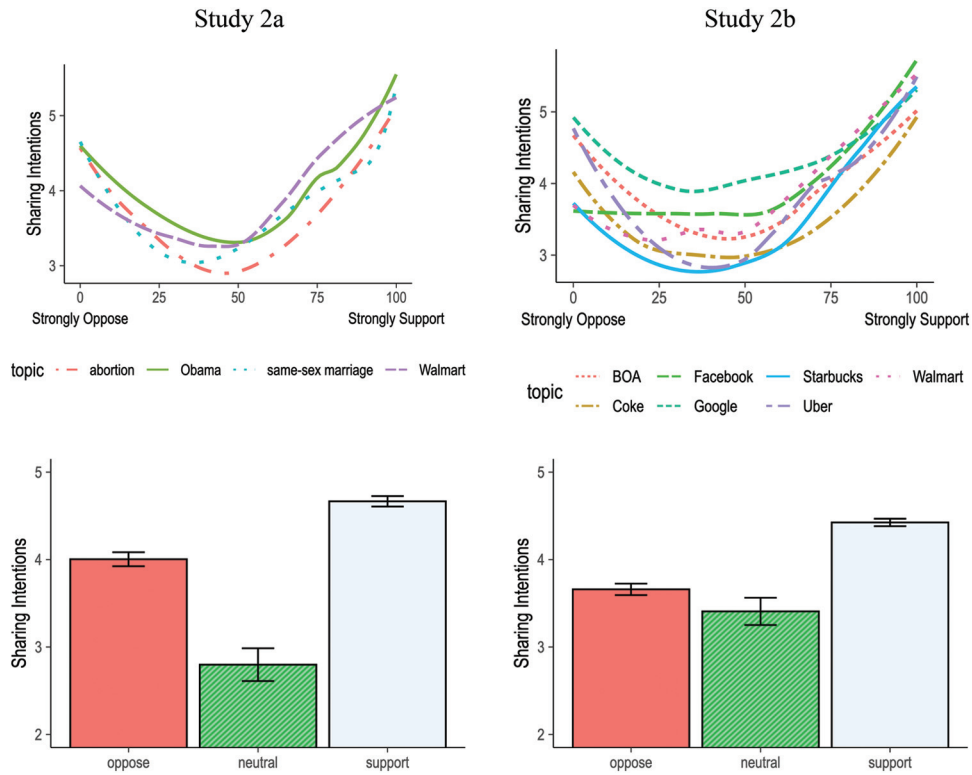
Figure 1
Relationship Between Support Versus Opposition and Sharing Intentions



Note. The top panels show the relationship on a continuous scale for Study 2a (left) and Study 2b (right). The bottom panels show the relationship by category for Study 2a (left) and Study 2b (right). Values below 50 were considered oppose, above 50 were considered support, and at 50 were considered neutral. Error bars represent standard errors. See the online article for the color version of this figure.

was framed in terms of the position they supported or the position they opposed.

After being randomly assigned to a framing condition, participants were asked to think about how they would approach their upcoming discussion with the other participant. They were told that if possible, they would be paired with someone who disagreed with them on the target issue. Then, participants completed measures indicating how likely they would be to share their view with the other participant if they turned out to be a friend or family member, an acquaintance, or a stranger, as in Study 2 (α = .84). To boost the realism of the study, participants were then given an opportunity to generate arguments and send them to the other participant (see online supplemental materials). The other participant was in fact a bot, which sent up to 10 preplanned counterattitudinal arguments, depending on how many arguments participants sent. When the chat was complete, participants completed demographic information and rated the importance of the gun control issue (1 = *extremely unimportant,* 7 = *extremely important*).

Results and Discussion

Participants indicated that gun control was an important issue to them, and equally so across conditions (M_{support} = 6.12, M_{oppose} = 5.93, $t(203) = 1.24$, $p = .22$). More germane to our hypothesis, participants reported greater sharing intentions in the support-framing condition (M = 4.97, 95% CI [4.73, 5.20]) than in the oppose-framing condition (M = 4.51, 95% CI [4.19, 4.82]), $t(204) = 2.39$, $p = .02$, $d = .33$. In other words, we found the predicted effect of support-oppose framing on sharing; participants indicated that they would be more likely to share their attitude with another participant when that attitude was framed in terms of what they supported rather than what they opposed. Importantly, Study 3 demonstrated this effect after randomly assigning participants to think in support versus oppose terms about the same attitudes. Supporting the idea that more gun control is harmful is logically equivalent to opposing the idea that more gun control is beneficial, and vice versa. Thus, Study 3 provides causal evidence for a support-oppose framing effect on sharing and suggests that this effect occurs when individuals are anticipating an actual interaction about an important and controversial topic.

Study 4: Mediation

Study 4 had several goals. First, we sought to provide evidence for the value-expression and impression-management pathways driving the effect of support-oppose framing on sharing. To this end, we manipulated framing and measured participants' feelings of value expressiveness and their perceptions of how a message recipient would view them if they shared their attitudes. Second, we examined an alternative account for the framing effect observed in

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Study 3. In Study 3, prior to the framing manipulation participants were asked to indicate which statement best characterized their attitude—in essence, which position they supported. Could it be that by focusing participants on their supported position, we inadvertently primed them to be more support-focused, which drove the effect of support framing on sharing? It seems unlikely that this methodological detail was responsible for the effect given that Studies 1 and 2 produced similar results despite very different methods. Nevertheless, to address this issue in Study 4, we asked participants to indicate both which statement they supported and which statement they opposed. Third, to situate this study in the context of a highly important and personally relevant topic, for which sharing decisions could have critical consequences for public health, we asked participants about their attitudes toward self-isolation in response to Coronavirus (COVID-19), a topic receiving global media attention at the time of the study.

Finally, Study 4 examined the possibility that the effect of support-oppose framing on sharing was confined to contexts in which people considered sharing with disagreeing others (i.e., a disagreeing audience). In Studies 2 and 3, our measure of sharing intentions specifically focused participants on disagreeing others so that we could speak to sharing dynamics in potentially divisive contexts. Moreover, based on our proposed mechanisms, it seems reasonable to believe that the effects could be limited to a disagreeing audience. For example, it is possible that disagreeing others make one's own values more salient and provoke an elevated drive to express those values, resulting in greater sharing. Likewise, perhaps when interacting with disagreeing others, impression management is particularly important and individuals seek to avoid offending their audience. On the other hand, it is also reasonable to believe that the desires to express one's values and be liked are strong enough to operate regardless of the audience's position on the topic. To gauge whether the support-oppose framing effect is restricted to the context of a salient disagreeing audience, in Study 4 we removed information about the sharing target's attitude. Given the inherent importance of value-expression and impression-management goals, we expected the effect to be robust to this modification.

Method

Based on a pilot test indicating a Cohen's *d* of .21 for our smallest effect, we set a target sample size of 1,000 participants, yielding over 90% power to detect our effect. In order to obtain 1,000 participants with complete data, we set a target sample size of 1,300 participants. Those who indicated that they had no opinion on the target issue completed a filler task and did not take part in the main study, leaving 1,211 participants with complete data. After exclusions specified in our preregistration,[8] 1,141 participants recruited from Prolific Academic remained in the final sample (53.29% female, M_{age} = 34.38).

At the outset of the study, participants were asked to indicate their attitude toward the target issue: self-isolation in response to the COVID-19 outbreak. After reading a description of the self-isolation recommendations, participants were given two statements: (1) It's extremely important that all Americans self-isolate (avoid public spaces) right now, and (2) It's not important that all Americans self-isolate (avoid public spaces) right now (see OSF for complete instructions). Participants were asked to use a drag-and-drop interface to move each statement to one of three categories that best characterized their position. Each category was indicated with a labeled box ("support," "oppose," and "neither"), and only one statement could be placed in each box. Thus, a participant who supported Statement 1 and opposed Statement 2 would drag Statement 1 to the support-labeled box and Statement 2 to the oppose-labeled box. The aim of this task was to lead all participants to consider both the position they supported and the position they opposed at the beginning of the study. Participants who supported one statement but did not oppose the other (or vice versa) completed a filler task. This design ensured that all participants in the oppose (support) condition actually considered themselves opposed to (in support of) the target statement.

Participants were then randomly assigned to a support-framing or oppose-framing condition and were told that they would answer questions about the statement, "It's extremely important that all Americans self-isolate (avoid public spaces) right now," or "It's not important that all Americans self-isolate (avoid public spaces) right now," depending on their original attitude and condition assignment. As in the previous experiment, all participants answered questions about an attitude position that was consistent with their overall attitude. This manipulation was designed to lead participants to think about their attitude in terms of what they supported or what they opposed.

Next, we asked participants to reflect on their attitude toward the appropriate statement and report their likelihood of sharing their views on this statement. The sharing index was the same as in the previous studies (three-item index, α = .88), the only difference being that we did not specify whether the target or recipient agreed or disagreed with participants' views. For example, we asked, "How likely would you be to share your views on this topic with a friend or family member?" without specifying that the friend or family member disagreed on this issue. We then included a series of measures designed to assess the value-expression and impression-management mechanisms. Value expressiveness was measured using two items: "How representative of your values is your attitude [in favor of/against] the statement, [statement]?" "To what extent would advocating [in favor of/against] the statement, [statement], reflect your values?" (r = .70, $p < .001$). To measure the impression participants thought sharing would elicit, we asked participants to imagine that they had decided to share their view and then indicate how positively they thought they would be viewed by the message recipient: "How positively would the message recipient view your values?" "How positively do you think the message recipient would view you as a person?" (r = .85, $p < .001$). Responses, provided on scales from 1 (*not at all*) to 7 (*extremely*), were averaged for each index. Finally, participants were asked to report their attitude toward the target statement (1 = *extremely opposed*; 7 = *extremely in favor*), along with demographic questions.

Results

We found a significant effect of support-oppose framing on sharing intentions. Participants reported greater sharing intentions when the issue was framed in terms that they supported (M = 5.97,

[8] https://aspredicted.org/blind.php?x=qi6g6p

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95% CI [5.89, 6.06]) rather than opposed (M = 5.29, 95% CI [5.15, 5.44]), $t(1139) = 8.08$, $p < .001$, $d = .48$). We also found a framing effect on the proposed mediators: value expressiveness and perceived impression. First, participants viewed their sharing as more value expressive in the support-framing (M = 6.13, 95% CI [6.05, 6.21]) rather than oppose-framing (M = 5.61, 95% CI [5.48, 5.74]) condition, $t(1139) = 6.73$, $p < .001$, $d = .40$. Participants also believed that the message recipient would have a more positive impression of them after sharing in the support-framing (M = 5.46, 95% CI [5.37, 5.54]) relative to oppose-framing (M = 4.87, 95% CI [4.74, 5.00]) condition, $t(1139) = 7.53$, $p < .001$, d = .44.

To determine whether the effect of condition on sharing intentions was driven by the proposed mediators, we used structural equation modeling to explore multiple pathways (Kline, 2005), using the lavaan package in R. We specified our hypothesized model using unidirectional paths to test two indirect pathways simultaneously. Thus, all reported effects control for the other pathway. In one pathway, support-oppose framing affected predicted value expressiveness, which predicted sharing intentions. In the other, support-oppose framing affected perceived impression, which also predicted sharing intentions. We first examined the value-expression pathway. Support framing led to greater value expressiveness, ß = .39, $Z = 6.31$, $p < .001$, which predicted greater sharing intentions, ß = .31, $Z = 8.08$, $p < .001$. There was a significant indirect effect through this pathway, $ab = .12$, Z = 5.63, $p < .001$. Second, we examined the impression-management pathway. Support framing led to more positive perceived impressions, ß = .44, $Z = 7.25$, $p < .001$, which predicted greater sharing intentions, ß = .21, $Z = 5.84$, $p < .001$. We found a significant indirect effect here as well, $ab = .09$, $Z = 4.66$, $p < .001$ (see Figure 2).

Discussion

In Study 4, we replicated the support-oppose framing effect on sharing using a large sample for a well-powered demonstration of the effect. We also demonstrated that this effect is not confined to contexts in which people consider sharing their views with a disagreeing audience. Moreover, Study 4 provided initial mechanism evidence. Participants saw support-framed attitudes as more expressive of their values than oppose-framed attitudes, which predicted relatively greater sharing intentions for support-framed attitudes. Participants also believed that sharing support-framed attitudes would elicit more favorable impressions from others, which boosted sharing intentions as well. In addition to providing mechanism evidence, Study 4 demonstrated that the framing effect occurs even when people explicitly reflect on both the position they support and the position they oppose before considering their likelihood of sharing.

Study 5: Moderation by Desired Impression

Studies 1–4 show that across a wide range of topics, support-oppose framing influences sharing intentions. Study 4 suggests that this effect occurs through two pathways: value expression and impression management. As outlined earlier, these mechanisms are rooted in interaction goals. People approach social interactions with different goals in mind, and two fundamental goals involve value expression and impression management. Based on our framework and proposed mechanisms, we would expect the effect of support-oppose framing to vary depending on the specific goals people focus on in any given context. If people cared less about impression management, for instance, we might expect the effect to be attenuated relative to what we have observed thus far.

To explore this possibility, Study 5 assessed the effect of support-oppose framing on sharing intentions in a context in which people's impression-management goals should be dampened. Specifically, we tested the effect of framing on sharing with dissociative outgroups—that is, reference groups people wish to avoid being associated with (Englis & Solomon, 1995; Turner, 1991; White & Dahl, 2006, 2007). Extant research suggests that people tend to avoid behaviors that they believe would link them to dissociative outgroups (Berger & Heath, 2008; Berger & Rand, 2008; Dunn et al., 2013; Ferraro et al., 2009; White & Dahl, 2006, 2007), and past research suggests that people change their reported attitudes in order to distance themselves from individuals with whom they do not want to be associated (Cooper & Jones, 1969). However, to our knowledge no research has explored people's sharing intentions with respect to dissociative outgroups.

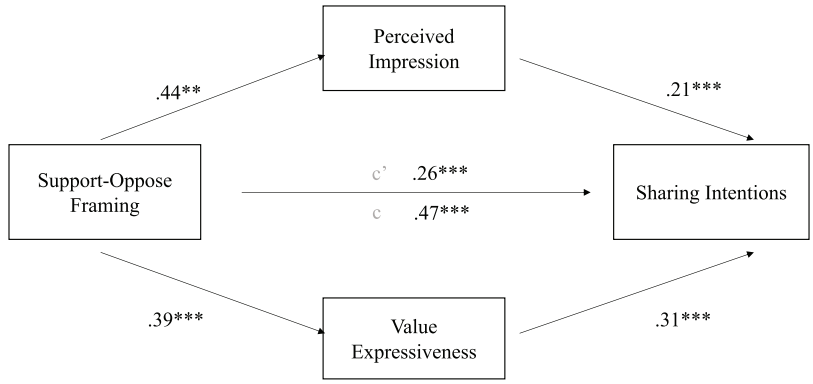
We posit that in addition to avoiding dissociative groups, individuals might also disregard their typical impression-management goals. This outcome might be especially likely when these groups are viewed as having different values or ideologies, given the centrality of values to an individual's identity and self-concept (Baumeister, 1995). Perhaps when considering groups with starkly different values, people are less concerned with impression management. Extending this reasoning to the current concerns, perhaps when people consider dissociative-outgroup targets, it attenuates the effect of support-oppose framing on sharing, due to the effect's partial basis in impression-management goals. Study 5 tests this possibility.

Method

As in Study 4, we set a target sample size of 500 participants per cell (2,000 participants total). After exclusions specified in our preregistration,[9] 1,701 Amazon Mechanical Turk workers remained in the dataset for analysis (56.67% female, M_{age} = 36.86). At the outset, participants were assigned to one of two outgroup target conditions: dissociative outgroup and control outgroup. In the *dissociative outgroup* condition, participants were asked to identify a group with values that are different from their own and with whom they wish to avoid being associated (adapted from White & Dahl, 2007). In the *control outgroup* condition, participants were asked to identify a group that they are not a part of, but would have no problem being associated with, that has some similar values to them and some different values from them. After participants typed in an appropriate outgroup, the experiment proceeded in much the same way as Study 4. In this case, participants read about Dick's Sporting Goods having recently implemented a policy limiting their gun sales. Participants indicated their attitude by selecting the statement that best characterized their view: "Dick's Sporting Goods' new gun policy (limiting sales) is a positive change," "Dick's Sporting Goods' new gun policy (limiting

[9] http://aspredicted.org/blind.php?x=9my4sm

Figure 2
Mediation Model for the Effect of Support-Oppose Framing on Sharing Intentions Through Perceived Impression and Value Expressiveness



Note. The notation c indicates the total effect of framing on sharing intentions; c' indicates the direct effect after controlling for the mediators. Coefficients presented are standardized linear regression coefficients. ** $p < .01$. *** $p < .001$.

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sales) is a negative change," or "no opinion." Participants who had no opinion were ineligible for the survey.

Participants were then randomly assigned to the support- or oppose-framing condition and told that they would answer questions about the statement, "Dick's Sporting Goods' new gun policy (limiting sales) is a [positive/negative] change," depending on their original attitude and condition assignment. To ensure that participants understood that their sharing could oppose the statement, we also included an indication of their attitude in the prompt. For example, participants who believed the change was negative and were assigned to the oppose-framing condition read: "We are interested in your thoughts on the idea that 'Dick's Sporting Goods' new gun policy (limiting sales) is a positive change', which you said you oppose." Participants who believed the change was negative and were assigned to the support-framing condition read: "We are interested in your thoughts on the idea that 'Dick's Sporting Goods' new gun policy (limiting sales) is a negative change', which you said you support."

After reading the statement, participants reported their sharing intentions. However, rather than indicating general sharing intentions, participants were asked to report their sharing intentions toward individuals in the identified outgroup. The same three-item index was used as in the previous studies, but we modified it to indicate the appropriate sharing target (e.g., "How likely would you be to share your views on this statement with a friend or family member who is a member of the group, [previously identified outgroup]?"; $\alpha = .88$). Thus, participants reported their sharing intentions with respect to a member of a dissociative outgroup or a control outgroup. Following the sharing items, participants answered questions on value expressiveness ($r = .69$, $p < .001$) and perceived impression ($r = .85$, $p < .001$), using the same measures as Study 4, as well as attitude certainty. Responses, provided on scales from 1 (*not at all*) to 7 (*extremely*), were averaged for each index. For a discussion of the effect of framing on attitude certainty and an analysis of certainty data in Study 5, see online supplemental materials.

Finally, participants completed manipulation checks assessing impression management concerns. Specifically, we assessed their desired impression after interacting with a member of the identified group, using two items: "How positively do you want this individual to view [your values/you as a person]?" Responses, provided on a scale from 1 (*extremely negatively*) to 7 (*extremely positively*), were averaged ($r = .73$, $p < .001$). In addition, participants were instructed to, "Imagine that you are choosing one of three messages to send to this individual: One that will make them like you, one that will make them dislike you, or one that will make them neutral toward you. Which would you send?," and given a trinary choice: "the one that would make them [like/dislike] me," "the one that would make them neutral toward me."

Results

The results from both manipulation checks suggested that participants cared less about impression-management goals when interacting with dissociative outgroups than control outgroups. Participants desired to be viewed more positively by control outgroup members ($M = 5.57$, 95% CI [5.50, 5.64]) than by dissociative outgroup members ($M = 4.90$, 95% CI [4.81, 4.99]), $t(1699) = 11.81$, $p < .001$. In addition, while 65% of individuals in the control outgroup condition indicated that they would send a message that would make them liked, only 33% of individuals in the dissociative outgroup condition indicated that they would send a message that would make them liked, $\chi^2(2, N = 1701) = 191.99$, $p < .001$.

Next, we assessed the effects of the framing and outgroup target manipulations on sharing intentions. Support framing led to greater sharing ($M = 3.93$, 95% CI [3.81, 4.05]) than oppose framing ($M = 3.49$, 95% CI [3.37, 3.62]), $F(1, 1699) = 24.41$, $p < .001$). There was also a significant effect of outgroup target on sharing intentions, such that individuals wanted to share less when the target was a member of a dissociative outgroup ($M = 3.04$, 95% CI [2.92, 3.16]) than when the target was a member of a control outgroup ($M = 4.40$, 95% CI [4.28, 4.51]), $F(1, 1699) = 267.68$, $p < .001$. However, these effects were qualified by a marginal interaction, $F(1, 1697) = 3.19$, $p = .07$ (see Figure 3, top panel). As predicted, the effect of support-oppose framing on

sharing intentions was greater in the control outgroup condition, $F(1, 1697) = 20.04$, $p < .001$, than in the dissociative outgroup condition, $F(1, 1697) = 3.68$, $p = .06$.

We also analyzed the effects of outgroup target and support-oppose framing on our mediators: perceived impression and value expressiveness. For perceived impression, the results paralleled the sharing intentions outcome. There was a significant effect of framing, $F(1, 1699) = 29.57$, $p < .001$, and of outgroup target, $F(1, 1699) = 557.01$, $p < .001$, but these effects were qualified by an interaction, $F(1, 1697) = 26.86$, $p < .001$ (see Figure 3, bottom left panel). Support framing led to more positive impression ratings in the control outgroup condition, $F(1, 1697) = 53.58$, $p < .001$, but not the dissociative outgroup condition, $F(1, 1697) = .002$, $p = .96$. For value expressiveness, there was a significant effect of framing, $F(1, 1699) = 243.79$, $p < .001$, and of outgroup target, $F(1, 1699) = 19.15$, $p < .001$. Interestingly, these effects were qualified by a marginally significant interaction, $F(1, 1697) = 2.86$, $p = .09$ (see Figure 3, bottom right panel), suggesting that the framing effect was greater in the dissociative outgroup condition, $F(1, 1697) = 146.31$, $p < .001$, than in the control outgroup condition, $F(1, 1697) = 53.58$, $p < .001$, though it was significant for both.

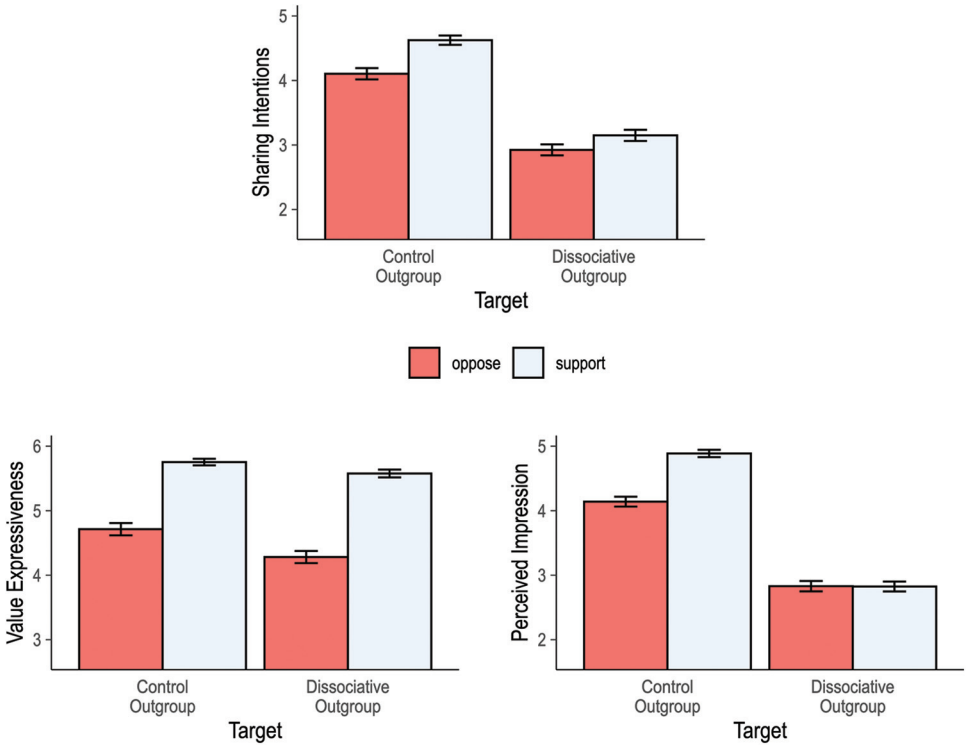
We performed a moderated mediation analysis testing whether the effect of support-oppose framing on sharing was mediated by our two proposed pathways and whether the link between framing and perceived impression was moderated by outgroup target. According to our theorizing, in the impression-management pathway, the path from support-oppose framing (independent variable) to sharing intentions (dependent variable) should operate through perceived impression, and therefore should apply in the control but not dissociative outgroup condition. To assess this model, we used lavaan in R and followed bootstrapping recommendations (Preacher et al., 2007; see Figure 4 for full model).

The first pathway tested value expression. Support-oppose framing affected value expressiveness, ß = .71, $Z = 14.52$, $p < .001$, which predicted sharing, ß = .19, $Z = 7.90$, $p < .001$, $ab = .14$, $Z = 6.99$, $p < .001$. The second pathway tested impression management. The first leg of this pathway, which regressed participants' perceived impressions on framing condition, outgroup condition, and the Framing × Outgroup target interaction yielded a significant interaction, ß = −.43, $Z = -5.13$, $p < .001$, suggesting that the effect of support-oppose framing on perceived impression depended on target. Support (vs. oppose) framing led to a more positive perceived impression from the control outgroup, ß = .43, $Z = 7.88$, $p < .001$, but not the dissociative outgroup, ß = −.003, $Z = -.04$, $p = .97$. The second leg of this pathway showed that perceived impressions predicted sharing, ß = .44, $Z = 19.77$, $p < .001$. The moderated mediation for this pathway was significant, index of moderated mediation = −.19, $Z = -4.94$, $p < .001$.

Discussion

In Study 5, we replicated the support-oppose framing effect and its mechanisms using a different target issue and provided

Figure 3
Interaction Between Target Outgroup and Support-Oppose Framing on Sharing, Perceived Impression, and Value Expressiveness



Note. Error bars show standard errors. See the online article for the color version of this figure.

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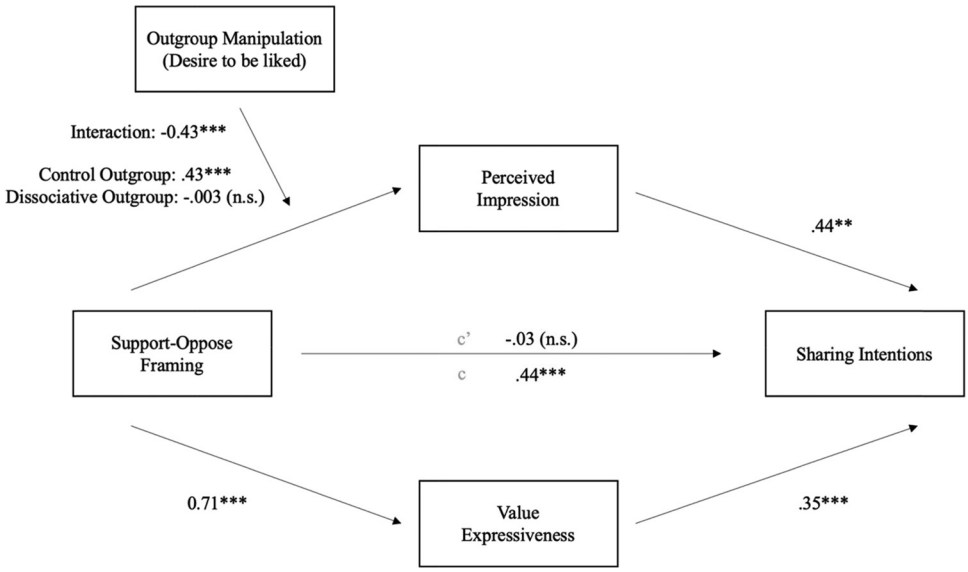
Figure 4

Moderated Mediation Model for the Effect of Support-Oppose Framing on Sharing Intentions Through Perceived Impression and Value Expressiveness, as Moderated by Outgroup Target



Note. The notation c indicates the total effect of framing on sharing intentions; c' indicates the direct effect after controlling for the mediators. Coefficients presented are standardized linear regression coefficients. ** $p < .01$. *** $p < .001$.

additional process insight through moderation. We found that the effect was attenuated when impression-management goals were relaxed. Specifically, when individuals considered sharing with a member of a dissociative outgroup—a group whose impression of them mattered less—the effect of support- versus oppose-framed attitudes was reduced. This attenuation was attributable to the impression-management pathway, such that the effect of framing on perceived impression was moderated by outgroup target. This finding underscored the role of impression-management goals in driving the support-oppose framing effect. When participants cared less about evoking a positive impression, the effect of framing on sharing intentions was diminished. It is also worth noting that the value-expression pathway remained intact in the dissociative-outgroup context, suggesting that value-expression motives affect sharing even with dissociative outgroups. This finding also rules out a potential alternative account that perhaps participants simply cared less about dissociative outgroups, which led to a decrease in processing or attention. If true, this inattention would lead to an attenuation of the framing effect across all pathways, including value expression. Thus, it seems that the observed attenuation is not driven by attentional differences, but rather by differences in impression-management motives.

General Discussion

Six studies reveal that people are more likely to share their views when their attitudes are framed in terms of what they support rather than what they oppose. This effect is separable from valence; it occurs for both positive and negative attitudes (Studies 1–2), and even for ambivalent attitudes (Study 1). Furthermore, the framing effect is causal: Reframing the same attitude in terms of what an individual supports (rather than opposes) leads to greater sharing, even when support- and oppose-framed attitudes are logically equivalent (Studies 3–5). This effect is driven by two pathways (Studies 4–5). In the first, people feel that support-framed attitudes are more value expressive, and in turn are more likely to share them. In the second, individuals believe that support framing will lead to a more positive impression, which contributes to greater sharing as well. This latter pathway, and the overall framing effect, is attenuated when people are less motivated to be liked, as in the case of dissociative outgroups (Study 5).

The support-oppose framing effect was robust to procedural variations. It emerged whether framing was measured (Studies 1–2) or manipulated (Studies 3–5), and across settings including in the field (Study 1), the lab (Study 3), and multiple online samples (Study 2, Studies 4–5). We also found the framing effect across a diverse array of social and political topics, including controversial social issues (e.g., same-sex marriage and gun control), current events (e.g., self-isolation in response to Coronavirus), politicians (e.g., Obama), and brands (e.g., Walmart and Starbucks). We replicated our findings regardless of whether the sharing items asked about targets who disagreed (Studies 2–3) or whose attitudes were unknown (Studies 4–5).

Finally, we used different methods of eliciting what participants supported versus opposed. In Studies 2a and 2b, participants indicated support versus opposition on a sliding scale. In Studies 3 and 5 (see also Supplemental Studies 1, 3, and 4), participants indicated which of two diametrically opposed statements reflected their view and then were randomly assigned the statement they supported or the statement they opposed. In Study 4 (see also Supplemental Study 2), participants categorized two diametrically opposed statements as the ones they supported, opposed, or neither supported nor opposed, before being randomly assigned to the support or oppose condition. Regardless of how support versus

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opposition was elicited, we observed the same result: support framing led to greater sharing intentions than oppose framing. This invariance suggests that the effect is not due to focusing participants on their supported position and does not depend on the elicitation method employed.

As noted earlier, we view the current research as making several critical contributions to our understanding of attitudes. First, it deepens our insight into the factors that drive attitude sharing and advocacy—topics of increasing import in attitudes and persuasion research (Tormala & Rucker, in press). In addition, it extends prior work on attitude functions and interaction goals to uncover some of the underlying motives that drive attitude expression. Finally, it underscores the importance of support-oppose framing (independent of attitude valence, a dominant focus in attitudes research) as an important construct, and opens the door to new research exploring the causes and consequences construing one's attitude in terms of support or opposition.

Alternative Accounts

Our studies provide consistent evidence for the predicted effect of support-oppose framing on sharing decisions and detail two psychological mechanisms driving this effect. By offering demonstrations of these effects across numerous settings and using varied procedures, we can eliminate some alternative accounts for our findings. Here, we highlight alternative explanations that we have yet to address and introduce evidence that helps rule them out.

Attitude Valence

First, as reviewed already, there is some evidence from past research that positive attitudes are associated with greater sharing-related outcomes than negative attitudes (e.g., Zunick et al., 2017). Could it be that attitude valence accounts for increased sharing under support- rather than oppose-framing conditions? To address this question, Study 2 used multiple issues to separate attitude valence from framing as predictors of sharing. We found that framing predicted sharing intentions, whereas valence did not. This study provided initial evidence for the notion that the valence effect in past research might be driven by, or at least linked to, support-oppose framing—that is, framing might account for prior findings related to valence and sharing, rather than vice versa. Unfortunately, Studies 3–5 (see also Supplemental Studies 1–4) focused on single issues and we were unable to conduct comparable analyses, because attitude valence in those studies was tied to specific attitude positions. For example, if people who had positive attitudes toward same-sex marriage were more likely to share their views, this could be about positive valence in general or it could be something unique about being positive toward same-sex marriage.

To explore this issue further, we conducted a meta-analysis of Studies 2–4, Supplemental Studies 1–4, and the Study 5 control condition. This analysis tested the effects of support-oppose framing and valence across multiple issues such that positive and negative valence would not always map onto the same positions (McShane & Böckenholt, 2017). Specifically, we examined three contrasts. First, to parallel past research, we compared the support-framing/positive attitude combination to the oppose-framing/negative attitude combination. The results were consistent with prior work: Support/positive attitudes led to greater sharing than did oppose/negative attitudes (estimate = .86, SE = .13, Z = 6.61, $p <$ .001). Second, to test for a framing effect, we compared the two support-framing combinations (support/positive and support/negative) to the two oppose-framing combinations (oppose/positive and oppose/negative). Support-framed attitudes were more likely to be shared than oppose-framed attitudes, regardless of valence (estimate = 1.35, SE = .17, Z = 7.94, $p <$ .001). Third, to test valence, we compared the two positive attitude combinations (support/positive and oppose/positive) to the two negative attitude combinations (support/negative and oppose/negative). Positive attitudes were more likely to be shared than negative attitudes (estimate = .38, SE = .18, Z = 2.11, p = .03), regardless of framing. Importantly, though, this valence effect was smaller in magnitude than the framing effect revealed in the second contrast. These results suggest that support-oppose framing and valence affect sharing independently, but framing may play a larger role. This finding suggests that support-oppose framing might provide an alternative explanation for effects that have been attributed to valence in prior work. This possibility warrants attention in future research.

Certainty and Consensus

In addition to valence, past research has identified attitude certainty and perceived attitude consensus as determinants of sharing (Cheatham & Tormala, 2015, 2017; Krosnick et al., 1993; see Tormala & Rucker, 2018). Could it be that support framing increased attitude certainty and/or boosted perceptions of the consensus surrounding one's view and these factors fostered greater sharing? Based on prior research, it seems unlikely. As reviewed earlier, Bizer et al. (2011) found that people hold oppose- rather than support-framed attitudes with greater certainty. We find that support- rather than oppose-framed attitudes trigger greater sharing—the opposite of what would be predicted based on a certainty or consensus account. Nevertheless, we assessed the certainty and consensus accounts in supplemental studies.

First, we assessed the role of certainty in Supplemental Study 1. In this study, participants reported their attitudes toward gun control before being randomly assigned to support- or oppose-framing conditions as in Studies 3–5. Following the manipulation, participants reported sharing intentions, value expressiveness, and perceived impression, followed by a measure of attitude certainty. Importantly, we found no difference in attitude certainty between the support- and oppose-framing conditions. Moreover, controlling for certainty left the mediating pathways through value expressiveness and perceived impression intact.

Supplemental Study 3 (which used a similar design with the same measures but changed the issue to same-sex marriage) did uncover a certainty effect, such that certainty was greater in the support- rather than oppose-framing condition (cf. Bizer et al., 2011). Again, though, controlling for certainty, the other mediating pathways remained significant. Also, compared with value expressiveness and perceived impression, certainty was the weakest contributor to the overall sharing effect. Together, Supplemental Studies 1 and 3 suggest that certainty might play a partial, inconsistent role in the framing effect on sharing, but it cannot account for the roles of value expression and impression management (see also Study 5 supplemental analysis).

Supplemental Study 2 assessed the role of perceived consensus. Here, participants indicated their attitude toward the target issue

(the Starbucks plastic straw ban) using the drag-and-drop interface from Study 4, were randomly assigned to support- or oppose-framing conditions, and then reported sharing intentions, value expressiveness, perceived impression, and perceived consensus ("If you had to guess, what percent of people agree with you [i.e., have a similar attitude to you] in your attitude toward this statement?"). Most pertinent, the effect of support-oppose framing on perceived consensus was not significant, and controlling for perceived consensus left the other mediating pathways intact. Thus, perceived consensus does not appear to be a primary driver of the observed effect.

Fluency

Does fluency play a role? Perhaps people find it easier to think about their attitudes in support than oppose terms, which triggers greater sharing in the former than latter case. For example, it could be that oppose framing is disfluent when it involves a double negative (or negates a negative; e.g., opposing the notion that immigration is bad), which is difficult to process. Importantly, though, we find that oppose framing is associated with reduced sharing regardless of whether it creates a double negative. Opposing that something is good, for instance, reduces sharing in the same way as opposing that something is bad, despite the fact only one of those positions negates a negative. As another source of disfluency, perhaps oppose frames require that people reflect on both an attitude and an "oppose" tag, which makes oppose-framed attitudes more difficult to retrieve and share. This process might resemble the notion of "false" tags (Gilbert, 1991; Gilbert et al., 1990), whereby false information is more difficult to retrieve and process than true information, because false information involves both the information itself and the "false" tag attached to it. In the current work, all statements or positions that participants consider sharing would be viewed as true. For example, despite the oppose frame, saying "I oppose the notion that immigration is good" is true if it reflects one's actual attitude. Thus, we do not believe that differential processing of truth versus falsehood underlies our findings.

In addition to conceptual grounds, there are empirical grounds for seeing fluency as an unlikely driver of the sharing effect. Prior work suggests that fluency boosts people's feelings of confidence and certainty (e.g., Haddock et al., 1999; Tormala et al., 2002; Tormala & Rucker, 2018). Thus, if fluency is a key driver of the effect of framing on sharing, we would expect support-framed attitudes to be held with greater certainty. As detailed already, the evidence for certainty is inconsistent. When we do see differences in certainty, certainty is the weakest contributor to the sharing effect, and as noted controlling for certainty leaves the proposed pathways intact. Thus, our evidence indirectly suggests that fluency is not crucial.

Moreover, even if fluency *did* prove important in the current context, it would not necessarily be at odds with the mechanisms we put forth. Most germane, we argue that one driver of the support-oppose framing effect is value expression: People see what they support, rather than oppose, as more representative of who they are (e.g., their core values). It makes sense, then, that people might also find it easier to think and talk about their attitudes in support terms. In other words, people might find it easier to think and talk about support-framed attitudes, because those attitudes feel more attuned to their core values. Indeed, perhaps the more value expressive an attitude feels, the more people think and talk about that attitude, which makes the attitude more accessible and, thus, fluent (see Barden & Tormala, 2014). In short, if fluency plays a role, it could be a part of the value-expression pathway we identified. Future research exploring fluency in this context would be worthwhile.

Informativeness

As another account, could it be that people view support as more informative than opposition about their underlying attitudes, and that differences in perceived informativeness guide sharing? Indeed, support frames might sometimes seem more informative than oppose frames. For example, in an election with multiple candidates (e.g., Candidates A, B, and C), knowing that someone supports Candidate A is more informative about where they stand and for whom they will vote than knowing that someone opposes Candidate B. In the context of the bipolar issues and attitudes explored in the current research, we surmise that this difference might be reduced such that supporting one side is more akin to opposing the other. Nevertheless, perhaps people extend their general perceptions of informativeness from some contexts to others, and perceived informativeness contributes to the effect of support-oppose framing on sharing. If so, we posit that perceived informativeness might be tied to the value-expressiveness pathway we uncovered in the current work. Indeed, the more value-expressive a particular position is, the more informative (e.g., about one's beliefs and values) it would seem to be.

To assess the role of informativeness, we conducted Supplemental Study 4. This study followed the same procedure as Supplemental Study 3, but added measures of informativeness (How informative is your attitude . . . about where you stand on this issue? To what extent would sharing your attitude . . . give another person clear information about where you stand on the issue?). We then explored multiple models to determine which best fit the data and found that perceptions of informativeness were closely tied to value expressiveness. Support framing increased perceptions of value expressiveness, which predicted greater perceived informativeness, which in turn boosted sharing intentions. In addition, the pathway from framing to sharing through perceived impression remained intact. Thus, initial evidence suggests that informativeness might be a consideration in sharing support- rather than oppose-framed attitudes, and it likely operates as part of the value-expression pathway. Further studies investigating the role of informativeness in sharing decisions would be valuable.

Priming

As a final account, does priming play a role in our effect? For instance, perhaps the words "support" and "oppose" prime agreeableness (or acquiescence) and disagreeableness (or resistance), respectively, and people are more likely to indicate that they would share after being primed to agree or acquiesce. To speak to this issue, we designed Supplemental Studies 1 and 3 such that support and opposition were implied as concepts without being directly stated in the study materials. In these studies, we asked participants to select the statement that best characterized their view, without mentioning support or opposition per se. Likewise, we told participants they would answer questions about the appropriate statement (which depended on condition assignment), without mentioning whether they supported or

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opposed that statement. Despite never mentioning "support" or "oppose," these studies replicated our main findings, making support- and oppose-priming an unlikely explanation for our findings.

Future Directions

Our focus in this research has been on documenting a support-oppose framing effect on sharing decisions and exploring the value-expression and impression-management mechanisms. As noted, in demonstrating a reliable tendency for support framing to stimulate greater sharing than oppose framing, our studies make several important contributions to the attitudes and persuasion literature. At the same time, this work raises a number of questions for future research. For instance, one next step would be to assess the amount or content of people's sharing (see supplemental studies for an initial exploration). Here we summarize other interesting directions.

Other Motives for Sharing

First, although we leveraged prior theorizing on attitude function and interaction goals to identify the value-expression and impression-management motives driving sharing, other motives exist and would be worth exploring in future studies. As one salient possibility, consider the motive to persuade, or change others' attitudes to bring them closer to one's own. Not surprisingly, past research reveals that persuasion goals can play an important role in attitudinal advocacy (Akhtar et al., 2013; Cheatham & Tormala, 2015, 2017). How would support-oppose framing affect persuasion intentions? How would the motive to persuade interact with the motives for value expression and impression management?

We speculate that support-oppose framing might affect persuasion intentions—that is, the specific intention to change others' attitudes—and that this effect could stem from the motives examined in the current studies. For instance, it could be that people express greater persuasion intentions when they think of their attitudes in support rather than oppose terms, because they believe that they will be more persuasive when they express their core values and elicit more favorable impressions. If true, persuasion intentions would operate like sharing intentions. In fact, as noted in Footnote 5, for exploratory purposes some of our studies did measure persuasion intentions and the results closely paralleled the sharing outcomes. Interestingly, though, it is also possible that support-oppose framing affects persuasion intentions, but through a different mechanism. For instance, maybe people believe support-framing will have greater efficacy than oppose-framing, because support-framing seems less adversarial or combative, which may increase receptiveness. In this case, people might have a motive to persuade, but framing affects the perceived efficacy of trying to do so. We see this general topic as ripe for future research.

Audience Factors

It would also be useful to explore audience factors and how they might moderate the effect of framing on sharing outcomes. Our studies revealed that people were more likely to share support- than oppose-framed attitudes with diverse audiences including friends, acquaintances, and strangers. We also observed this effect in contexts involving an audience who disagreed with participants (Studies 2–3) or whose attitudes were unknown (Studies 4–5).

Would the effect emerge in the context of an agreeing audience—that is, an audience with a similar attitude to the sharer? It is possible that when people consider sharing with an agreeing audience, impression-management motives are relaxed, which dampens the effect of framing on sharing decisions. Likewise, it might seem less important to express one's values when the audience already agrees, if for instance it feels like "preaching to the choir," which further undermines the framing effect. On the other hand, value-expression and impression-management motives might be so deeply rooted that they operate regardless of the audience's position on a topic. To examine this issue, in Supplemental Study 3 we directly manipulated whether participants focused on similar or dissimilar attitude targets (i.e., an agreeing or disagreeing audience). As detailed there, this factor did not influence sharing decisions. Thus, our initial evidence suggests that audience attitude does not moderate the present effect.

Nonetheless, audience characteristics are bound to matter. Indeed, Study 5 identified dissociative outgroups as a case in which impression-management motives are relaxed and the effect of support-oppose framing on sharing is attenuated. A natural next step would be to explore audience factors that might relax value-expression motives. As one possibility, the presence of some audiences (e.g., highly admired or aspirational individuals) might highlight discrepancies between people's actual (e.g., slightly favorable toward gun control) and desired (e.g., extremely favorable toward gun control) attitudes (see DeMarree et al., 2014). In such cases, people might feel some confusion about their values or might doubt that their actual attitudes reflect their true values. As a result, the value-expression motive might recede, thus attenuating the effect of support-oppose framing on sharing. On the other hand, it is possible that value expression is a central motive that is activated across audiences, making it an especially difficult pathway to moderate. More work in this area is needed.

Regulatory Focus

Are promotion and prevention goals triggered by support and oppose framing? Based on prior work on regulatory fit (e.g., Idson et al., 2000; Higgins, 2000; Sparks, 2020), there are several possibilities. One is that support framing feels like pursuing desired outcomes, thereby leading to a promotion focus, whereas oppose framing feels like avoiding an undesirable outcome, thereby leading to a prevention focus. If support-oppose framing does influence regulatory focus, the effect of framing on sharing might be strongest among promotion-oriented individuals, and weakest among prevention-oriented individuals. This is an interesting possibility worth testing directly in future studies.

In our view, support and oppose framing are distinct from promotion and prevention goals. Consider their conceptual relationship. In principle, support framing could be used to facilitate prevention goals and oppose framing could be used to facilitate promotion goals. For example, supporting a ban on guns, immigration, or gay marriage could be construed as prevention-focused if doing so feels like preventing undesired outcomes. Empirically too, our data are not consistent (in any obvious way) with the notion that these constructs are related. Indeed, if support (oppose) framing fostered a promotion (prevention) focus, we might expect a matching effect, such that support and oppose framing feels more natural, or right, for positive and negative attitudes, respectively. However, as discussed already, our studies provide no evidence for a matching effect between framing and valence. Nevertheless, the potential role of regulatory focus in guiding the effects of support- and oppose-framed attitudes—on sharing and other outcomes—would be worth exploring in subsequent work.

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Cultural Differences

In the current studies, we made an effort to test the support-oppose framing effect across diverse samples. We assessed it on Twitter (Study 1), with university community members in the lab (Study 3), and with diverse American samples on both Prolific Academic (Study 4) and Amazon Mechanical Turk (Studies 2, 5, Supplemental Studies 1–4). Across samples, we found consistent evidence that support framing leads to greater sharing than oppose framing. However, it is an open question whether this effect occurs cross-culturally. Past research suggests that what is considered normative to express differs across cultures. For example, compared with Chinese-American couples, European-American couples express more positive and less negative emotive behaviors during conflicts, a difference that has been attributed to cultural differences in expectations around emotional expression (Tsai et al., 2006). It is possible that norms around support versus opposition also differ cross-culturally. If individuals in some cultures believe that expressing opposition is more acceptable, the support-oppose framing effect may be attenuated, or even reversed. On the other hand, in cultures where expressing opposition is viewed as more counternormative than in American culture, it is likely that there would be an even greater difference in sharing between support- and oppose-framed attitudes. Further research is necessary to understand how cultural factors moderate the effect.

Conclusion

People have more opportunities to share their attitudes and opinions—to express their support or opposition—than ever before. The current research investigates whether thinking about attitudes in terms of support or opposition affects the inclination to share. Across studies, we find that support framing promotes greater sharing than does oppose framing, regardless of attitude valence. We provide converging evidence for this effect and its psychological mechanisms across ten studies, totaling over 70,000 observations, conducted in the field, in the lab, and online. Our findings have deep implications for attitudes research, where the focus has been on attitude valence rather than attitude framing, and for a rapidly expanding literature on attitude expression and advocacy behavior. In addition, our work contributes to research on framing effects by both introducing a new framing effect and identifying the mechanisms that drive it. Our findings also have potential implications for other domains, including disclosure in close relationships, by shedding light on when and why people share potentially controversial opinions. Indeed, understanding the factors that lead people to express their views, particularly to others who disagree, is increasingly pressing in multiple domains of psychology and in the real world. The current work represents a potentially important advance in this regard.

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